# Markdown and README.md

Quick guide to documenting Python projects in VS Code and GitHub

**A .md file is plain text with a lightweight formatting syntax. On GitHub, README.md usually acts as the technical front page of a project.**

## 1. What is a .md file?

The **.md** extension stands for **Markdown**. It is a lightweight text format: you write normal text with a few symbols, and tools such as VS Code or GitHub render headings, lists, code, links, and tables.

It is not a special binary format. You can open and edit it like any text file, and it remains readable even without rendering.

```
# Ballistic Flight Simulator v1.0

## Description
Simple vertical motion simulator.

## Inputs
- Initial altitude
- Initial velocity
- Time step
```

## 2. How to create README.md in VS Code

With your project folder open in VS Code, create a new file inside the mini-project folder using the Explorer panel.

• Right-click the project folder.

• Select **New File**.

• Type exactly **README.md**.

• Open the file and start writing Markdown.

For example, your project can look like this:

```
ballistic_flight_simulator_v1/
|-- ballistic_flight_simulator_v1.py
|-- README.md
`-- notes/
```

## 3. Creating it from the terminal

If you are already inside the project folder, you can create the file directly from the terminal.

## Linux / WSL:

```
touch README.md
```

## PowerShell:

```
New-Item README.md
```

## 4. Basic Markdown syntax

Markdown uses a few simple symbols. The following are enough for almost all of our early README files.

| Goal | Markdown | Result |
|---|---|---|
| Title | # Title | Main heading |
| Section | ## Section | Subheading |
| Bold | **text** | highlighted text |
| Inline code | `dt` | variable name |
| List | - item | bullet list |
| Code | ```python ... ``` | code block |

## 5. Previewing Markdown in VS Code

VS Code can render Markdown for you. Open **README.md** and press **Ctrl + Shift + V** to open the preview. This lets you write the raw Markdown and check how it will look on GitHub.

```
## Inputs

- `h0`: initial altitude [m]
- `v0`: initial velocity [m/s]
- `dt`: integration time step [s]
```

**Tip: the README is not the code. It is the technical explanation that lets someone understand the project before opening the .py files.**

## 6. What should a good README explain?

• **What** the project does.

• The **physical model** and main equations.

• The **numerical method** used.

• **Inputs** and units.

• **Outputs** produced.

• **Validation** against a reference.

• Current **limitations**.

• **How to run** the project.

# 7. Template for Ballistic Flight Simulator v1.0

For our first mini-project, the README does not need to be long. A structure like this is enough:

```
# Ballistic Flight Simulator v1.0

## Description

## Physical Model

## Numerical Method

## Inputs

## Outputs

## Validation

## Limitations

## How to Run
```

## Example initial content

```
# Ballistic Flight Simulator v1.0

## Description
Vertical flight simulator implemented in Python.

## Physical Model
The model assumes constant gravity and no aerodynamic drag.

## Numerical Method
The equations are integrated using the Explicit Euler method.

## How to Run
python ballistic_flight_simulator_v1.py
```

# 8. Where to store it

```
Python_Engineering/
`-- 01_python_bootcamp/
    `-- week_01_logic/
        `-- class_01_equation_to_algorithm/
            `-- mini_project/
                `-- ballistic_flight_simulator_v1/
                    |-- ballistic_flight_simulator_v1.py
                    |-- README.md
                    `-- notes/
```

## 9. Why does it matter in a portfolio?

When someone opens a GitHub repository, the README is usually the first thing they see. A good README quickly communicates the goal, technical level, and how to reproduce the project. Good documentation is therefore part of software engineering, not an optional extra.

> **Code: solves the problem.**
> **README: explains what problem is solved, how it works, and how to use it.**

## Learning Tracker

| Concept | Learned |
|---|---|
| .md file | Yes |
| Basic Markdown | Yes |
| README.md | Yes |
| VS Code preview | Yes |
| Project documentation | Introduction |
| GitHub preparation | Introduction |